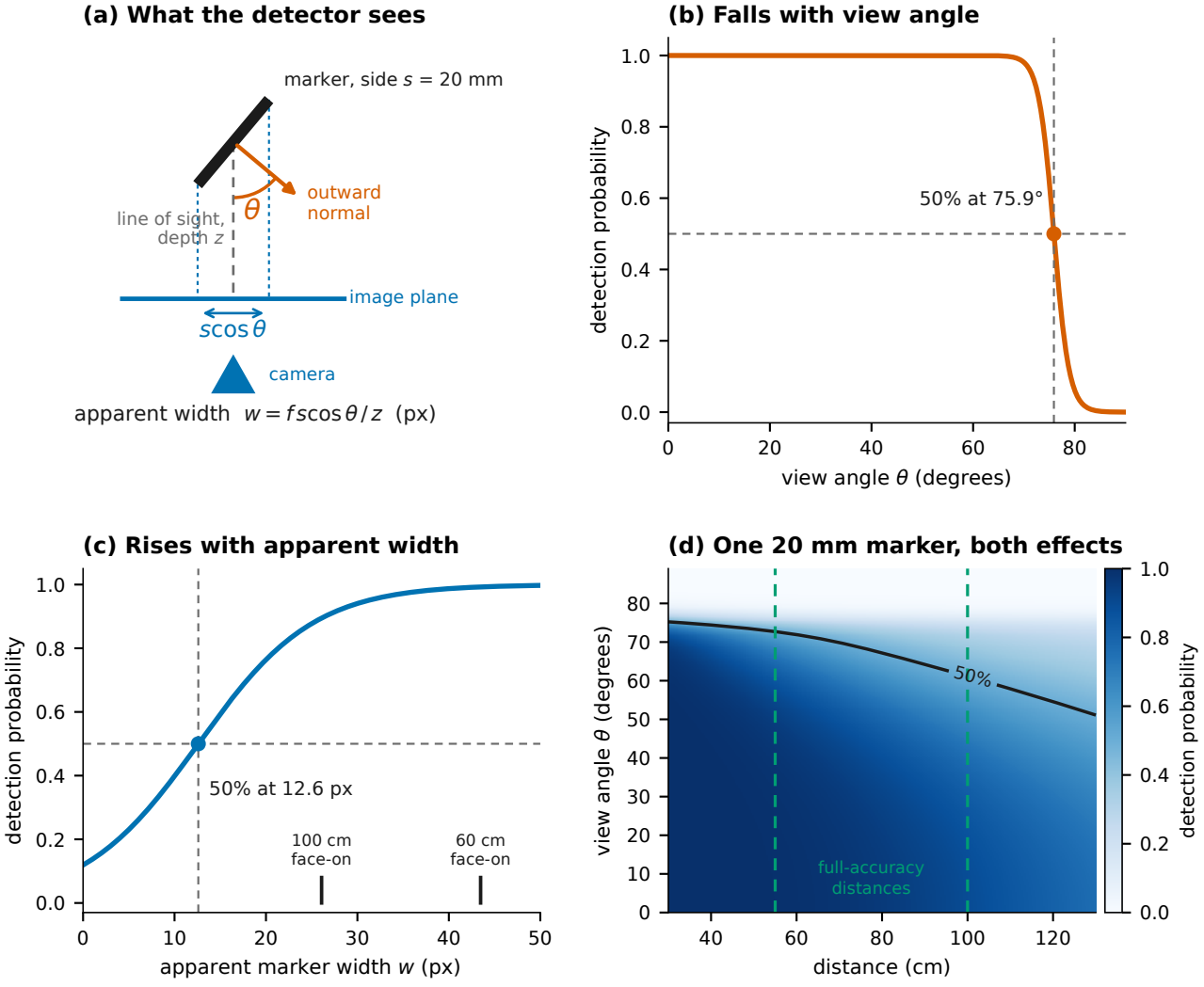

(a) What the detector sees
marker, side s = 20 mm
outward
normal
θ
line of sight,
depth z
image plane
s cos θ
camera
apparent width w = f s cos θ / z (px)
(b) Falls with view angle
detection probability
1.0
0.8
0.6
0.4
0.2
0.0
50% at 75.9°
0
20
40
60
80
view angle θ (degrees)
(c) Rises with apparent width
detection probability
50% at 12.6 px
100 cm
face-on
60 cm
face-on
0
10
20
30
40
50
apparent marker width w (px)
(d) One 20 mm marker, both effects
view angle θ (degrees)
80
70
60
50
40
30
20
10
0
50%
full-accuracy
distances
40
60
80
100
120
distance (cm)
detection probability